

Midterm exam reference questions

Chapter 1

1 、 Questions

1. Describe the main functions of the operating system.
2. Please explain the three major design purposes of the operating system.
3. What are the 4 main functions of the operating system?
4. What are the main design considerations for personal computer operating systems?
5. What are the design considerations for the operating system on a mainframe computer and the operating system on a personal computer?
6. Describe the advantages of a handheld system and the limitations in terms of hardware.
7. What are the advantages of "microkernels"?
8. What is the main job of a CPU? How many tasks can a CPU perform at once?
9. What is a register? What are the types?
10. In a multiprocessor system, if all processors are equivalent (no master-slave relationship), what do we call it?
11. Please explain how interrupts, exception interrupts, and traps differ from each and give examples of each.

2 、 Advanced questions

1. When will the multi-program system and the time-sharing system switch schedules?
2. Try to analyze the role of user mode and system mode from the perspective of system security and efficiency.
3. Why can't the user's program switch to system mode by itself?

Chapter 2

1.Question

1. Explain the difference between the process and the program ◦
2. Please describe the main functions and differences of the various schedulers ◦

3. Please describe which statuses are usually available for process(trips) ?
4. Describe the state in which the process leaves the running state in those cases
5. Describe the circumstances under which a new itinerary is generated
6. Please explain the possible reasons for the end of the tour ◦
7. Briefly describe common queues in the operating system and explain their purpose ◦
8. Please explain what body switching is ?
9. Body switching is an additional burden for the system, what factors affect the time it takes to switch contexts? ◦
10. When a thread is created, what resources are used? What's the difference with creating an itinerary ?
11. Why does the body switch of a thread faster than the body of a process?
12. Please explain the relationship between CPU burst, I/O burst and CPU-dominated and I/O main processes?
13. Define preemptible and non-front-running scheduling methods, and give examples of situations where front-running scheduling methods must be used?
14. In a multi-tiered queue scheduling approach, what are the advantages of having different time slices for different queues?
15. Assume a scheduling method that is most beneficial for the most recent time that uses the least CPU time. Why this scheduling method is beneficial for I/O-focused trips?

2. Calculation Problems

1. There is a set of itineraries as follows :

<u>Itinerary</u>	<u>CPU burst time (ms)</u>
P ₁	15
P ₂	7
P ₃	4
P ₄	10

The arrival time is 0 and is in the order of P₁, P₂, P₃ and P₄ ◦

Use FCFS and non-preemptive SJF scheduling methods to find the average wait time, respectively ◦